

Linux 100 — A Cheatsheet for Complete Beginners

100 commands every new Linux user actually reaches for, grouped by what you're trying to do. Each entry gives you the command, what it does, and a realistic example. Nothing exotic — this is the stuff you'll type in your first weeks on a terminal.

1. Navigation & Basics

1. **pwd** — print the full path of the directory you're currently in — `pwd`
2. **ls** — list the contents of a directory — `ls`
3. **ls -la** — list everything, including hidden files, in long format — `ls -la`
4. **cd** — change directory — `cd Documents`
5. **cd ..** — move up one directory level — `cd ..`
6. **cd ~** — jump straight to your home directory — `cd ~`
7. **cd -** — jump back to the previous directory you were in — `cd -`
8. **tree** — show a directory's contents as a visual tree — `tree Documents`
9. **clear** — clear the terminal screen — `clear`
10. **exit** — close the current terminal session — `exit`

2. File & Directory Management

1. **touch** — create an empty file (or update its timestamp) — `touch notes.txt`
2. **mkdir** — create a new directory — `mkdir project`
3. **mkdir -p** — create nested directories in one go — `mkdir -p project/src/utils`
4. **cp** — copy a file — `cp notes.txt notes_backup.txt`
5. **cp -r** — copy a directory and everything inside it — `cp -r project project_backup`
6. **mv** — move or rename a file/directory — `mv notes.txt archive/notes.txt`
7. **rm** — delete a file — `rm notes_backup.txt`

8. **rm -r** — delete a directory and its contents — `rm -r old_project`
9. **rm -rf** — force-delete without confirmation (use with real caution) — `rm -rf temp_folder`
10. **rmdir** — delete an empty directory — `rmdir empty_folder`

3. Viewing & Editing Files

1. **cat** — print a file's entire contents to the screen — `cat notes.txt`
2. **less** — scroll through a file one page at a time — `less big_log.txt`
3. **head** — show the first 10 lines of a file — `head server.log`
4. **tail** — show the last 10 lines of a file — `tail server.log`
5. **tail -f** — watch a file live as new lines are appended (great for logs) — `tail -f server.log`
6. **nano** — beginner-friendly terminal text editor — `nano notes.txt`
7. **vim** — powerful (steeper learning curve) terminal text editor — `vim notes.txt`
8. **file** — tell you what type of file something actually is — `file mystery_file`
9. **wc** — count lines, words and bytes in a file — `wc -l notes.txt`
10. **diff** — show the differences between two files — `diff old.txt new.txt`

4. Permissions & Ownership

1. **ls -l** — view file permissions, owner and group — `ls -l notes.txt`
2. **chmod +x** — make a file executable — `chmod +x install.sh`
3. **chmod 755** — set exact permissions using octal notation — `chmod 755 script.sh`
4. **chmod -R** — apply a permission change recursively — `chmod -R 755 project/`
5. **chown** — change who owns a file — `sudo chown alex notes.txt`
6. **chown user:group** — change owner and group at once — `sudo chown alex:staff notes.txt`
7. **chgrp** — change a file's group only — `sudo chgrp staff notes.txt`
8. **umask** — view or set the default permissions new files get — `umask`
9. **sudo** — run a single command with administrator privileges — `sudo apt update`

10. **su** — switch to another user account — `su - alex`

5. Searching & Finding

1. **find -name** — search for files by name under a directory — `find . -name "*.txt"`
2. **find -type** — search by item type (file or directory) — `find . -type d`
3. **find -mtime** — find files modified within a time window — `find . -mtime -1`
4. **locate** — very fast filename search using a prebuilt index — `locate notes.txt`
5. **updatedb** — refresh the index that locate searches — `sudo updatedb`
6. **which** — show the full path of a command — `which python3`
7. **whereis** — locate a binary, its source and its manual page — `whereis ls`
8. **grep** — search for text inside a file — `grep "error" server.log`
9. **grep -r** — search for text recursively through a directory — `grep -r "TODO" project/`
10. **grep -i** — case-insensitive text search — `grep -i "warning" server.log`

6. Processes & System Monitoring

1. **ps** — show currently running processes — `ps`
2. **ps aux** — show a detailed list of every process on the system — `ps aux`
3. **top** — live, auto-refreshing view of running processes — `top`
4. **htop** — a friendlier, colour-coded version of top (may need installing) — `htop`
5. **kill** — stop a process using its process ID — `kill 4821`
6. **kill -9** — force-stop a process that won't quit normally — `kill -9 4821`
7. **killall** — stop all processes matching a name — `killall firefox`
8. **jobs** — list jobs running in the current shell session — `jobs`
9. **bg / fg** — send a job to the background, or bring it to the foreground — `bg` / `fg`
10. **uptime** — show how long the system has been running and its load — `uptime`

7. Networking

1. **ping** — test whether a host is reachable — `ping google.com`
2. **ip a** — show network interfaces and their IP addresses — `ip a`
3. **ifconfig** — the older tool for the same job as ip a — `ifconfig`
4. **curl** — fetch or send data to/from a URL — `curl https://example.com`
5. **wget** — download a file from the internet — `wget https://example.com/file.zip`
6. **ssh** — securely log in to a remote machine — `ssh alex@192.168.1.10`
7. **scp** — securely copy files to/from a remote machine — `scp notes.txt alex@192.168.1.10:/home/alex/`
8. **ss -tulnp** — show which ports are open and listening — `ss -tulnp`
9. **traceroute** — show the network path packets take to a host — `traceroute google.com`
10. **hostname -I** — show this machine's own IP address(es) — `hostname -I`

8. Package Management

1. **apt update** — refresh the list of available packages (Debian/Ubuntu) — `sudo apt update`
2. **apt upgrade** — upgrade all installed packages to their latest version — `sudo apt upgrade`
3. **apt install** — install a new package — `sudo apt install git`
4. **apt remove** — uninstall a package — `sudo apt remove git`
5. **apt search** — search for a package by name/keyword — `apt search text editor`
6. **dpkg -l** — list every package currently installed — `dpkg -l`
7. **dnf install** — install a package (Fedora / RHEL) — `sudo dnf install git`
8. **yum install** — install a package (older RHEL / CentOS) — `sudo yum install git`
9. **pacman -S** — install a package (Arch Linux) — `sudo pacman -S git`
10. **snap install** — install a self-contained snap package — `sudo snap install code`

9. Users, Groups & Access

1. **whoami** — print the current logged-in username — `whoami`
2. **id** — show the current user's ID and group memberships — `id`
3. **adduser** — create a new user account (friendlier, interactive) — `sudo adduser alex`
4. **useradd** — create a new user account (lower-level) — `sudo useradd alex`
5. **passwd** — set or change a user's password — `sudo passwd alex`
6. **usermod -aG** — add a user to an additional group — `sudo usermod -aG sudo alex`
7. **deluser** — remove a user account — `sudo deluser alex`
8. **groupadd** — create a new group — `sudo groupadd developers`
9. **groups** — show which groups the current user belongs to — `groups`
10. **last** — show recent login history for the system — `last`

10. Archives, Compression & Disk

1. **tar -cvf** — bundle files into a .tar archive — `tar -cvf backup.tar Documents/`
2. **tar -xvf** — extract a .tar archive — `tar -xvf backup.tar`
3. **tar -czvf** — bundle and compress files into a .tar.gz archive — `tar -czvf backup.tar.gz Documents/`
4. **tar -xzvf** — extract a compressed .tar.gz archive — `tar -xzvf backup.tar.gz`
5. **zip -r** — compress a folder into a .zip archive — `zip -r backup.zip Documents/`
6. **unzip** — extract a .zip archive — `unzip backup.zip`
7. **df -h** — show free/used disk space in human-readable form — `df -h`
8. **du -sh** — show the total size of a directory — `du -sh Documents/`
9. **free -h** — show memory (RAM) usage in human-readable form — `free -h`
10. **lsblk** — list disks and partitions attached to the system — `lsblk`

*A note before this becomes a PDF: `rm -rf`, `chmod -R`, and anything run with **sudo** are the items on this list that can actually hurt you if typed carelessly. Worth a one-line callout box in the final design.*